Memoryless MAC Channels — Results

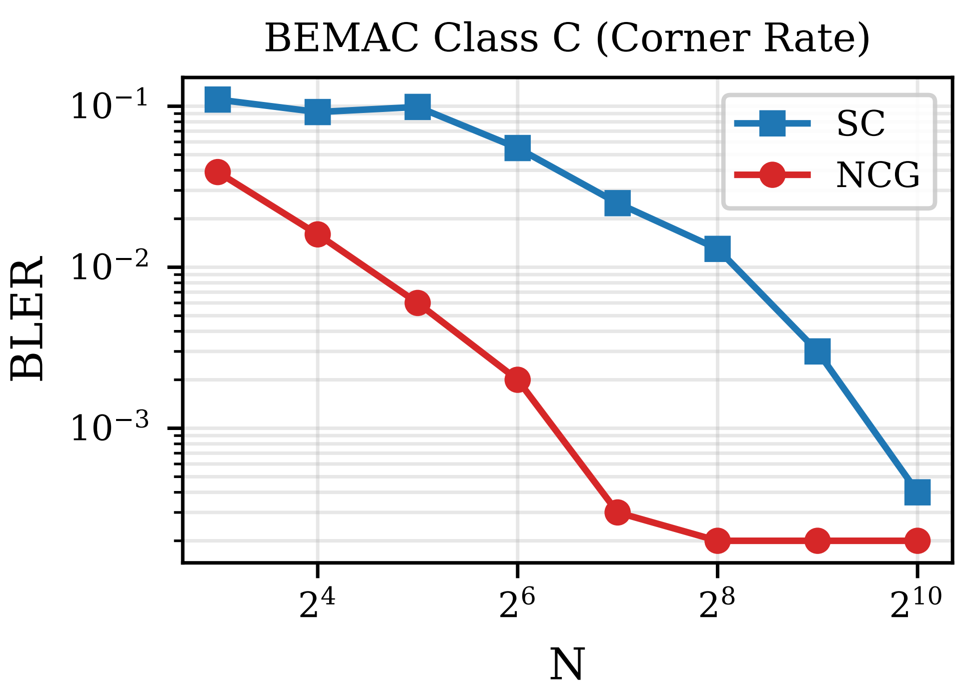
Neural Polar Decoders for Two-User MAC

April 2026

1 BEMAC Class C (Corner Rate)

$Z = X + Y, \quad Z \in \{0, 1, 2\}$ (deterministic).
Path: `make_path(N, N)`, $R_U \approx 0.30$, $R_V \approx 0.60$.
Decoder: NCG (d=16, h=64).

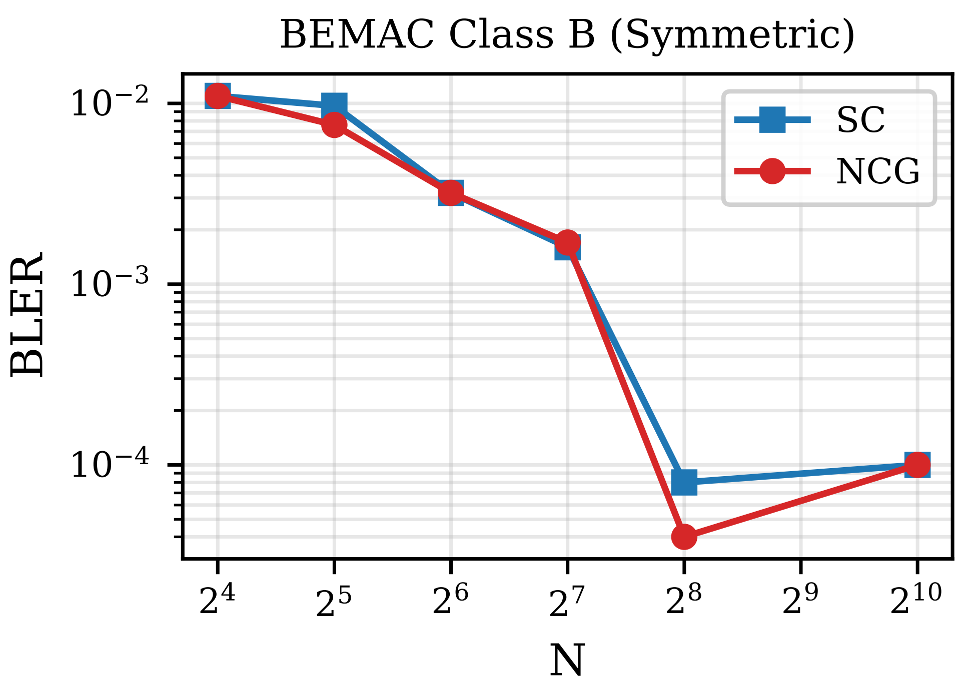
N	k_u/k_v	SC	NCG	**NCG/SC**	CW
8	2 / 5	0.110	**0.039**	**0.36**×	3K
16	5 / 10	0.092	**0.016**	**0.18**×	3K
32	10 / 19	0.099	**0.006**	**0.06**×	3K
64	19 / 38	0.055	**0.002**	**0.03**×	3K
128	38 / 77	0.025	**0.0003**	**0.01**×	20K
256	77 / 154	0.013	**0.0002**	**0.01**×	50K
512	154/307	0.003	**0.0002**	**0.06**×	50K
1024	307/614	0.0004	**0.0002**	0.42×	64K



2 BEMAC Class B (Symmetric Rate)

$Z = X + Y$. Path: `make_path(N, N/2)`, $R_U \approx 0.50$, $R_V \approx 0.70$.
Decoder: NCG (d=16, h=64).

N	k_u/k_v	SC	NCG	**NCG/SC**	CW
16	8 / 11	0.011	0.011	1.08×	5K
32	16 / 22	0.0097	**0.0076**	**0.78**×	15K
64	32 / 44	0.0032	0.0032	1.00×	40–48K
128	64 / 89	0.0014	0.0017	1.21×	50–90K
256	128/178	8×10^{-5}	4×10^{-5}	0.50×	50K
512	256/358	0.0	0.0	—	2K
1024	512/716	10^{-4}	10^{-4}	1.00×	10K



3 GMAC Class C (Corner Rate)

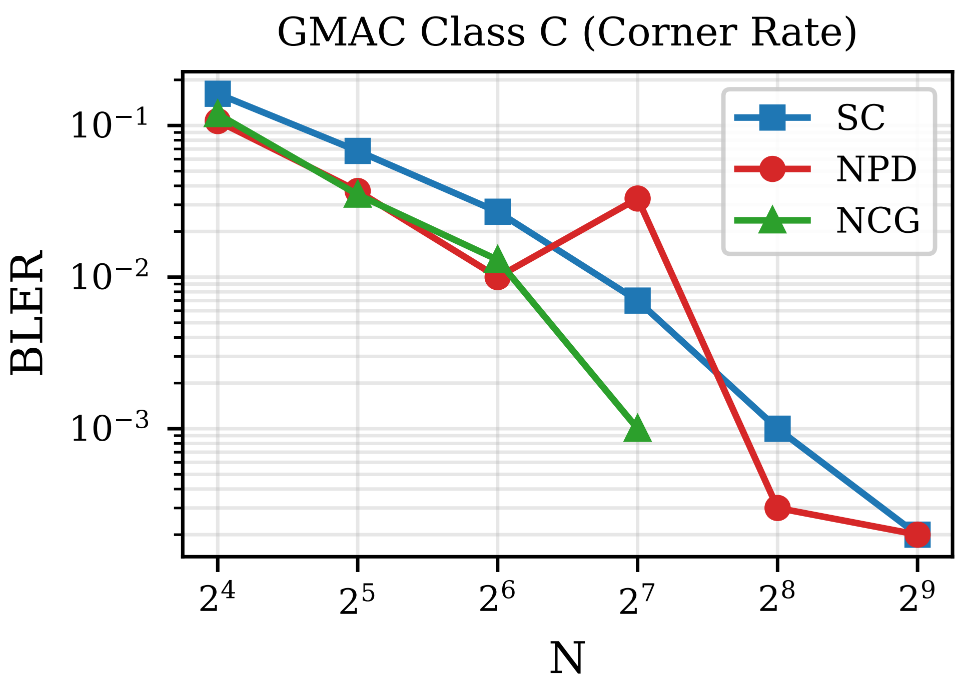
$Z = (1-2X) + (1-2Y) + W, \quad W \sim \mathcal{N}(0, \sigma^2)$, SNR $= 6\,$dB.
Path: `make_path(N, N)`, $R_U \approx 0.23$, $R_V \approx 0.45$.
NPD/NCG use MI-designed frozen sets (co-adapted with the neural decoder).

4 GMAC Class B (Symmetric Rate)

$Z = (1-2X) + (1-2Y) + W$, SNR $= 6\,$dB.
Path: `make_path(N, N/2)`, $R_U = R_V \approx 0.48$.
Decoder: NCG (d=16, h=64, sequential training).

N	k_u/k_v	SC	NPD	**NPD/SC**	NCG	CW
16	4 / 7	0.162	**0.107**	**0.66**$\times$	0.119	10K
32	7 / 15	0.068	**0.037**	**0.55**$\times$	0.035	10K
64	15 / 29	0.027	**0.010**	**0.37**$\times$	0.013	10K
128	30 / 58	0.007	0.033	4.7$\times$	**0.001**	10–20K
256	59/117	0.001	3×10^{-4}	0.27$\times^*$	—	20–50K
512	119/233	4×10^{-4}	2×10^{-4}	0.50$\times^*$	—	50K

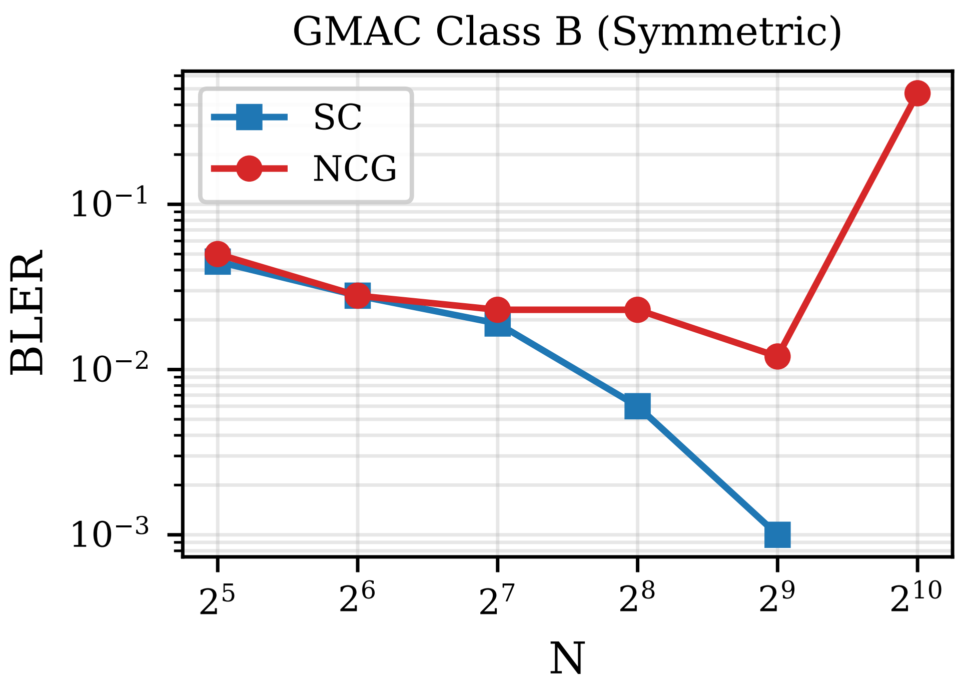
*N=256: 6 errors/20K CW. N=512: 11 errors/50K CW. Both unreliable.



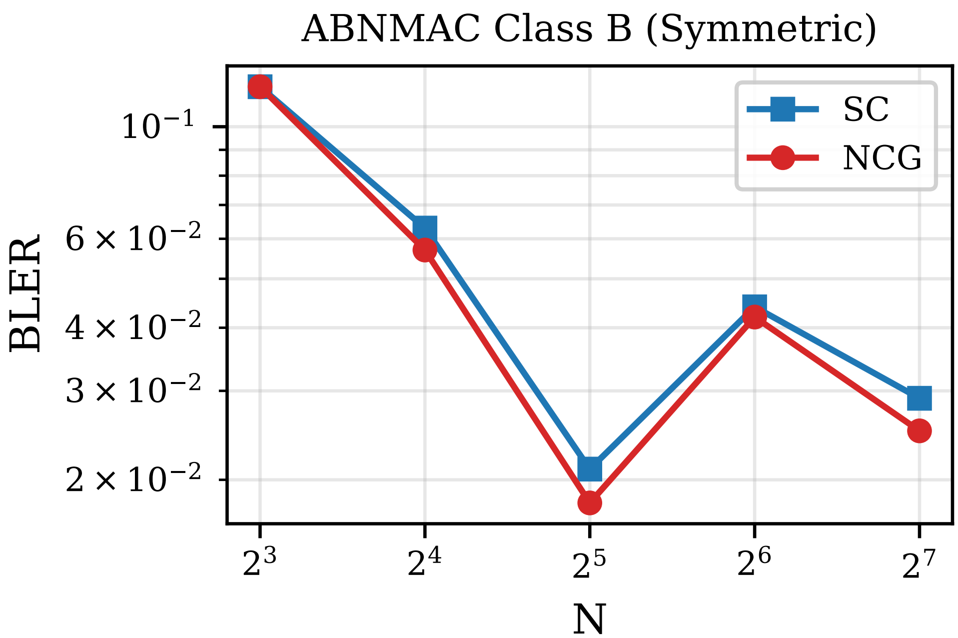
5 ABNMAC Class B (Symmetric Rate)

$Z = (X \oplus E_x,\ Y \oplus E_y)$, correlated binary noise.
Path: `make_path(N, N/2)`, $R_U = R_V \approx 0.30$.
Decoder: NCG (d=16, h=64).

N	k_u=k_v	SC	NCG	**NCG/SC**	CW
32	15	0.045	0.050	1.12×	3K
64	31	0.028	0.028	1.02×	5K
128	62	0.019	0.023	1.23×	6K
256	123	0.005	0.023	**4.9**×	5–35K
512	246	∼0.001	0.012	12×	10K
1024	491	—	0.470	BROKEN	2K



N	k_u=k_v	SC	NCG	**NCG/SC**	CW
8	3	0.120	0.120	1.00×	10K
16	5	0.063	**0.057**	**0.91**×	10K
32	10	0.021	**0.018**	**0.85**×	10K
64	22	0.044	**0.042**	0.95×	5K
128	45	0.029	0.025	0.87×	4K


ABNMAC Class B (Symmetric)
SC
NCG
BLER
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
2^3
2^4
2^5
2^6
2^7
N